

SI No.	Code	Description	Unit	Revised	wbs	Reply
				QTY		
E		Insulation		0	insulation	
				0	insulation	
E.1		Supply, Installation Testing & Commissioning of Thermal Insulation (For Exposed Ductwork) for application using Class "0" Closed cell nitrile rubber elastomeric foam / Physically cross-linked polyolefin foam insulation laminated with chemically treated glass cloth (Color shall be approved by architect) of 7 mil / 0.18 mm thickness along with manufacturer recommended adhesive. Insulation material manufactured should have fire properties - Class 0 Rating as per BS 476 Part 7 & Part 6 test & HB and V0 Rating as per UL 94. Insulation shall have a minimum Thermal Conductivity of 0.035 W / m K @ 0 Deg C as per EN 12667/ EN ISO 8497. Insulation shall have a Water Vapor Diffusion Resistance Factor of 7,000 as per EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209. Insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and Fibre Free. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.		0	insulation	Complies
		13 mm thk	Sq.M	948	insulation	Noted
		19 mm thk	Sq.M	948	insulation	
		25 mm thk	Sq.M	0	insulation	
				0	insulation	
E.2		Supply, Installation Testing & Commissioning of Thermal Insulation (Outdoor Ducting) material shall be Closed Cell Elastomeric Nitrile Rubber / Physically cross-linked polyolefin foam with UV resistant metal finishing on one side. Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 0°C as per EN 12667. Water Absorption by Volume of insulation material shall be < 0.2% as per ASTM C 1763 / ASTM C 209. The base insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990. Density of base foam insulation material shall be between 40 to 60 Kg/m3. The insulation material shall be dust and fibre free. The insulation material shall be formaldehyde free. The insulation material shall be CFC & HCFC free. The insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of 0 Deg.C as per EN 14706. The base insulation material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero. System material is a multi layer laminate of aluminium, coated with a special UV protection and a PVC backing. Covering material shall be non-metallic. Covering material shall provide mechanical resistance with an excellent aesthetic look. Covering material shall be glossy finish. The Covering material shall be of 230 microns thickness as per EN 22286 and weight shall be 340 gsm as per EN 22286. Method A. Thickness of the insulation shall be as specified for the individual application. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.		0	insulation	Complies
		13 mm thk	Sq.M	0	insulation	Noted
		19 mm thk	Sq.M	0	insulation	
		25 mm thk	Sq.M	744	insulation	
				0	insulation	
E.3		Supply, Installation Testing & Commissioning of Thermal Insulation for chilled water piping application using Class "0" Closed cell nitrile rubber elastomeric foam insulation with a 3-layered non-metallic composite cladding of minimum 350 micron thickness / aluminium cladding along with manufacturer recommended adhesive where the VOC level should not exceed more than 850 grams / Litre. Insulation material manufactured should have fire properties - Class 0 Rating as per BS 476 Part 7 & Part 6 test & HB and V0 Rating as per UL 94. Insulation shall have a minimum Thermal Conductivity of 0.037 W / m K @ 20 Deg C as per EN 12667/ EN ISO 8497. Insulation shall have a Water Vapor Diffusion Resistance Factor of 7,000 as per EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209. Insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and Fibre Free. Density of insulation material should be 40-60 kg /m³. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.		0	insulation	Complies
		Note: For exposed chilled water pipes and pipes within shafts, Aluminium cladding should be considered.		0	insulation	Noted
		25mm thk	Sq.M	720	insulation	
		32mm thk	Sq.M	1440	insulation	
		50mm thk	Sq.M	2640	insulation	
				0	insulation	
E.4		Supply, Installation Testing & Commissioning of Thermal Insulation for refrigerant piping application using Class "0" Closed cell nitrile rubber elastomeric foam insulation laminated with chemically treated glass cloth (Grey/Blue/Red/Green/ Black in Colour) of 7 mil / 0.18 mm thickness along with manufacturer recommended adhesive. Insulation material manufactured should have fire properties - Class 0 Rating as per BS 476 Part 7 & Part 6 test & HB and V0 Rating as per UL 94. Insulation shall have a minimum Thermal Conductivity of 0.035 W / m K @ 0 Deg C as per EN 12667/ EN ISO 8497. Insulation shall have a Water Vapor Diffusion Resistance Factor of 7,000 as per EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209. Insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and Fibre Free. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.		0	insulation	Complies
		13mm thk	Sq.M	0	insulation	Noted
		19mm thk	Sq.M	0	insulation	
				0	insulation	
E.5		Supply, Installation Testing & Commissioning of Thermal Insulation for condensate piping application using Class "0" Closed cell nitrile rubber elastomeric foam insulation along with manufacturer recommended adhesive. Insulation material manufactured should have fire properties - Class 0 Rating as per BS 476 Part 7 & Part 6 test & HB and V0 Rating as per UL 94. Insulation shall have a minimum Thermal Conductivity of 0.035 W / m K @ 0 Deg C as per EN 12667/ EN ISO 8497. Insulation shall have a Water Vapor Diffusion Resistance Factor of 7,000 as per EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209. Insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and Fibre Free. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.		0	insulation	Complies
		6 mm thk	Sq.M	0	insulation	Noted
		9 mm thk	Sq.M	0	insulation	
				0	insulation	

